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Does pad displacement correlate to impact severity metrics in football helmets?

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Abstract

As efforts to reduce sport-related head injuries continue, there is a need to better understand the mechanisms of an impact that influence the risk of sustaining a concussion. This study presents a method which can be used to separate football helmet pad displacement into normal and rotational components. Finite element models of three popular football helmets were used to simulate a variety of impacts per the National Football League (NFL) helmet testing protocol. Pad displacement components were evaluated for correlation with impact severity metrics including Head Injury Criterion (HIC), Diffuse Axonal Multi-Axis General Evaluation (DAMAGE), and Head Acceleration Response Metric (HARM). When combining all locations, displacement was not found to correlate well with severity metrics ($R^2 < 0.44$). However, trends did exist with normal displacement when evaluating data by impact location. Normal displacement was found to correlate strongly with HARM, HIC, and Peak Resultant Linear Acceleration (PRLA) at the side upper (SU) location ($R^2 = 0.79$–0.8). Overall, it was observed that helmets which maximize normal displacement without bottoming out performed better. Tangential displacement was not found to correlate well with any impact severity metrics even when evaluated by location. This study has provided a method to evaluate components of helmet pad normal displacement during simulations which may help guide design and optimization efforts for helmets under impact in critical locations.

Keywords

FEA, impact, football helmets, displacement, HIC, DAMAGE, HARM, TBI, concussion, mTBI

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Introduction

Approximately 1.6–3.8 million traumatic brain injuries (TBIs) occur annually in the USA due to sports and recreational activities, roughly half of which go unreported.[1] Concussions are referred to as mild traumatic brain injury (mTBI), making up 75% of all TBIs.[2]

In any given season, 10% of collegiate and 20% of high school football players will experience a brain injury, with over 60% of concussions in high school athletes coming from football.[3] NFL data shows that the number of concussions in professional football rose from 126 in 2021 to 149 in 2022, an 18% increase. This rate was 14% higher than the previous 3-year average (2018–2020) of 130 concussions.[4] It is important to note that while concussions occur in football, they are also prevalent in many other sports and recreational activities, including cycling, basketball, soccer, and snow sports.[3,5]

Concussions can sometimes prove to be fatal. Except for 1990, there has been at least one football-related death each year from 1945 to 1999, 69% of which were the direct result of head injury.[6] Of the 42 football-related deaths in the U.S. between 2008 and 2015, 29 (69%) were caused by head injuries.[7] Partly due to the increased awareness of the incidence and consequences of concussion, the number of athletes of all ages participating in football decreased from 8.4 to 5.2 million from

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2006 to 2017.[8] In response to increased concussion awareness, the NFL began the Head Health TECH Initiative, which devoted $60 million to better understanding the biomechanics of head injuries as well as developing better protective equipment to prevent them.[9] In addition, the PAC-12, a collegiate athletics conference, donated $3.5 million from 2019 to 2020 to improve student-athlete health and well-being.[10]

In football, the helmet is arguably the most important piece of protective equipment. Helmets utilize various materials and padding designs intended to absorb impact energy and minimize the acceleration experienced by the head and brain. Generally, a football helmet consists of the outer shell which is often hardened plastic or polymer and surrounds all of the padding components, a facemask which is a rigid structure relative to the rest of the helmet and protects the user's face and a chin strap or chin cup which secures the helmet to the head underneath the chin. In addition, helmets incorporate a comfort liner which is often a softer foam that sits against the head underneath the shell and energy attenuation liner. This space encompassed by the structures between head and helmet shell is known as the offset.

The impact attenuation liner is a layer of pads or structures which are specifically designed to absorb energy from an impact and reduce its transfer to the head and brain. These liners vary significantly by helmet and utilize different deformation mechanics to achieve the same goal. For example, the Riddell helmet is a large offset helmet possessing a harder outer shell coupled with a Vinyl Nitrile (VN) foam energy attenuation liner. This is a relatively simple padding design which lends itself nicely to compressive analysis. In contrast, the Vicis helmet model studied has the largest offset, possesses a softer, deformable outer shell, and utilizes buckling or collapsable geometries in the energy attenuation liner. While markedly different in their approach to impact mitigation, each helmet's impact attenuation layer is designed to reduce the acceleration of the head resulting from an impact.

Head impacts result in linear and rotational accelerations, each of which can contribute to the probability of receiving a concussion. A study by Zhang et al.[11] found that linear accelerations over 82g (the acceleration due to gravity) or rotational accelerations over 6000 rad/s^2 result in a 50% probability of concussion. Because these acceleration values offer probabilities of injury and are not definitive, additional measures are necessary to determine whether an individual is concussed. Current methods of diagnosis are indirect and involve measures of balance, cognitive tests, and oculovestibular measures.[12] For example, the Sport Concussion Assessment Tool version 5 (SCAT5) uses measures of orientation, symptom monitoring, and neurocognitive function to evaluate injury. Currently, there is no conclusion about whether an athlete's SCAT5 results should be compared to a normative data set of a healthy population or to their own baseline results.[13] While no single test has absolute accuracy in evaluating concussions, current diagnostic tests are reasonably accurate when coupled with symptom monitoring.[14]

Considering the difficulty in identifying concussions, developing a set of protective equipment specifications to verify equipment safety is vital. The National Operating Committee on Standards for Athletic Equipment (NOCSAE) has established criteria for testing and certifying football helmets. Based on NOCSAE standards, a helmet must not exceed 1200 Severity Index (SI), a measure of linear acceleration, or 6000 rad/s^2 in laboratory testing.[15] Another impact metric, the Head Injury Criterion (HIC), is a function of resultant linear acceleration and time. The criterion was based on the biomechanics of the head during impact and was originally designed to assess crash safety in vehicles.[16]

Diffuse Axonal Multi-Axis General Evaluation (DAMAGE) is another head injury metric that measures the angular acceleration of the head to determine brain strain and has been shown to correlate well with finite element analysis results. DAMAGE considers both the direction of acceleration and the time duration of an impact in its calculation.[17] Moreover, the Head Acceleration Response Metric (HARM) is a linear weighted combination of both HIC and DAMAGE, which describes the response of a football helmet to an impact at a speed and location defined by NFL testing standards. The HARM calculation multiplies HIC and DAMAGE scores by coefficients predetermined from regression analysis of dummy impact reconstructions and adds them together. The Helmet Performance Score (HPS) was developed for NFL helmet performance rankings and is a weighted combination of 18 HARM scores. The 18 HARM scores correspond to three impact speeds at six locations per NFL helmet testing standards. The Methods section will provide a more detailed explanation of each metric.[17]

It is crucial to recognize that no one metric can accurately capture all aspects of helmet response during impact. For instance, HIC does not consider inertial loading, nor does it utilize the entire time history profile of an impact, thus, it does not always correlate well with experimental results.[18,19] Additionally, DAMAGE involves the solution of a second-order differential equation, meaning higher peak acceleration values can have lower DAMAGE scores contrary to what is often expected.[20] Finally, HPS offers only a single number to indicate helmet performance and does not offer any insight into the mechanics of an impact.

Given the difficulty in understanding and evaluating the mechanics of an impact, it is beneficial to use computational modeling to determine aspects of an impact that may not be feasible to do experimentally. Generally, researchers have used acceleration-based metrics in determining impact severity, including Levadnyi et al.,[21] who used computational modeling to study the effect of impacts on helmeted versus unhelmeted head models and assessed results using HIC.

Gabler et al.[22] investigated brain strain during impact to determine severity. Others have explored the effects of optimizing helmet components to reduce the effects of impact on the brain.[23,24] In addition, there have been attempts to exploit various impact-absorbing materials to improve helmet performance and athlete safety.[25] For example, the Vicis Zero1 utilizes a relatively soft outer shell combined with buckling column structures in lieu of traditional foams to reduce translational and rotational accelerations of the head.[26] The Xenith X2E model incorporates thin-walled, collapsible structures that dissipate energy by gradually venting air to offer a slower change in resulting velocity of the model.[27] However, a method of evaluating components of internal pad displacement experimentally remains elusive for various reasons, including short impact durations (typically less than 60 ms) and difficulties outfitting a helmet with adequate sensors to measure deformations.

Generally, the goal of helmet pad designs is to optimize the magnitude of pad displacement and time duration of an impact within the design space. There are two primary reasons for this. First, a pad will absorb energy until it has reached maximum displacement, whether in compression or shear. Once a pad has maximally displaced, it will no longer absorb energy and will transfer energy from an impact to the head, leading to elevated acceleration values. Given the acceleration-based formulation of most helmet impact severity metrics, increasing accelerations of an impact results in worse helmet performance scores. By optimizing the amount of displacement possible within a pad, the range of impacts for which the helmet may be effective increases. In addition, both HIC and DAMAGE metrics are based on acceleration-time history profiles for an impact. Because of the relationship between displacement and acceleration, increasing the duration of an impact may result in reduced impact severity metrics. Given the relationship of displacement to acceleration and acceleration to impact metrics, we hypothesize that pad displacement must inherently play a role in determining impact severity.

To the best of our knowledge, no prior attempts have evaluated the normal and tangential components of pad displacement during an impact and correlated them to helmet impact metrics. This paper seeks to investigate the correlation between displacement components and corresponding helmet metrics while increasing understanding of the mechanisms that influence an impact. First, we develop computer code that uses nodal coordinate data from an impact simulation to determine both linear and tangential displacement of the helmet pad. Next, we used acceleration data from the same impacts to calculate impact severity metrics (HIC, DAMAGE, and HARM). Finally, we investigate correlations between each component of displacement and the severity metrics for each impact. We hypothesize that helmets that exhibit larger pad displacement (i.e. "ride down") of the interior energy mitigation system without "bottoming out" (i.e. foam densification) will result in lower severity metric scores.

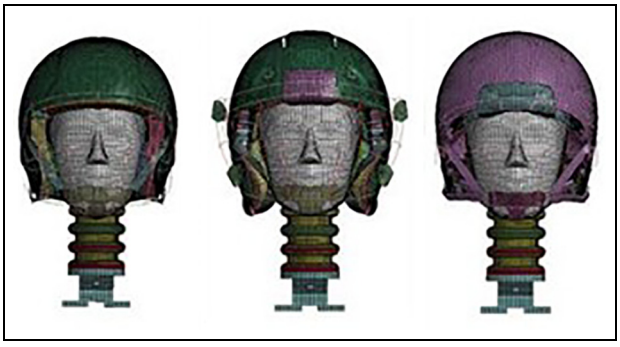

Figure 1. Exemplary image of Riddel helmet FE model (left), Vicis FE model (middle), and Xenith FE model (right).

Methods

Model and geometry development

The helmet finite element models used in this study were the 2016 Riddell Speed Classic, 2016 Xenith X2E, and 2017 Vicis Zero1 (Figure 1), which represent three unique helmet offsets and impact mitigation systems The Riddell model was developed by the KTH Royal Institute of Technology, the Xenith model was developed by the University of Waterloo, and the Vicis model was developed by the University of Virginia. All models are available to download for free on the BioCore LLC website, and each was validated by comparison to experimental laboratory impact results. A quantitative comparison of simulation and experimental results was performed using CORellation and Analysis (CORA). The Riddell and Xenith helmets received a CORA score of 0.79, while the Vicis helmet received a CORA score of 0.74.[28–30]

Each helmet model consists of several assemblies, including the linear impactor, Hybrid III (HIII) head and neck, the helmet, and the facemask. The linear impactor consists of the ram, backing plate, load cell, vinyl nitrile foam (VN600), and nylon end cap.[31]

The Hybrid III head-neck model was chosen to simulate the effects of an impact on a helmet-head assembly. The HIII assembly consisted of the rubber head skin, rigid skull component, head mount, and neck. The impactor and head-neck models were consistent for each helmet simulation.[31]

The helmet and facemask models were created by modifying existing CAD models to improve simulation efficiency. Materials for each helmet component were validated via experimental testing. Detailed information regarding model setup and material properties for each assembly can be found in the corresponding user manual provided by BioCore LLC.[28–30]

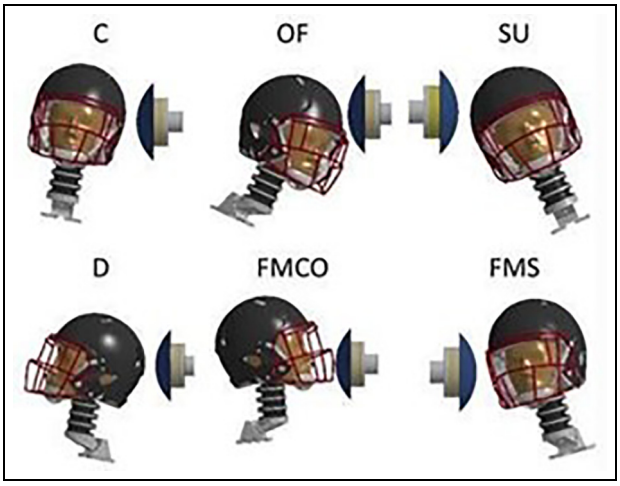

Figure 2. Impact locations used in NFL helmet test protocol.[14]

Simulation protocol

Simulations of each helmet model were conducted in LS-DYNA.[32] Each helmet was tested at six impact locations and three different speeds per location. The impact speed, impact location, helmet model orientation, and linear impactor orientation for each simulation were designed to match NFL testing protocols and were not altered in this study. Specifically, impact locations consisted of side (C), rear (D), side upper (SU), oblique front (OF), facemask side (FMS), and facemask center oblique (FMCO) and the impact speeds were 5.5, 7.4, and 9.3 m/s (Figure 2).[33] Detailed information regarding the setup and orientation of each model can be found in the corresponding helmet model user manual.[28–30]

Displacement calculation

MATLAB code was generated to determine both normal and tangential displacement components for each impact. To do this, time history data for three nodes were taken from the outer pad surface at the first timestep in which the linear impactor makes contact with the helmet. These nodes were taken from the center of the impact region using a visual inspection of a von Mises stress map at the impacting surface. Time history data for three nodes at the inner pad surface were also recorded.

Time history data for each simulation was truncated at 30 ms, as this corresponded to the time window in which maximum acceleration of the headform was observed to occur and with the duration in which HIC was calculated. Moreover, this removed any inaccurate data caused by an abnormality in the simulation, such as the helmet falling off the headform or the model intersecting itself at later time points. A detailed description of the code used is presented in the Supplemental Material to this manuscript.

Impact severity metrics

HIC is an impact metric based on resultant linear acceleration and time. For this study, HIC15 was used, which indicated the calculations were done using a 15-ms time window using the equation found in the Supplemental Material. DAMAGE is an impact metric that considers rotational acceleration time histories for an impact and is direction specific. HARM is a unitless value calculated from the weighted combination of HIC and DAMAGE using the equation found in the Supplemental Material.[16,17]

Correlation analysis

Once displacement time history data were obtained for all simulations, correlations between these data and helmet impact metrics (HIC, DAMAGE, and HARM) were evaluated. This was done by assessing the peak normal and tangential displacement values for each helmet combination, impact used to assess the correlation between normal displacement and each impact metric. This process was then repeated for the tangential component of displacement.

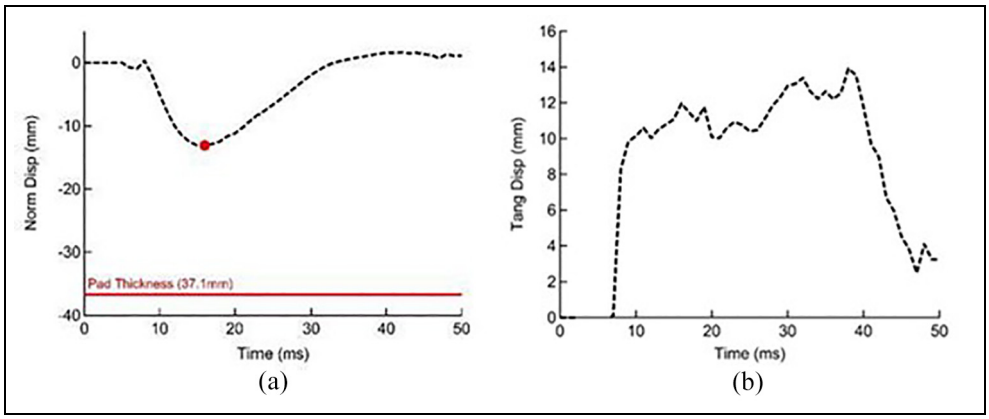

Figure 3. Representative time histories for: (a) normal displacement and (b) tangential displacement during an impact. The selected impact was at the rear location of the Riddell helmet at 5.5 m/s.

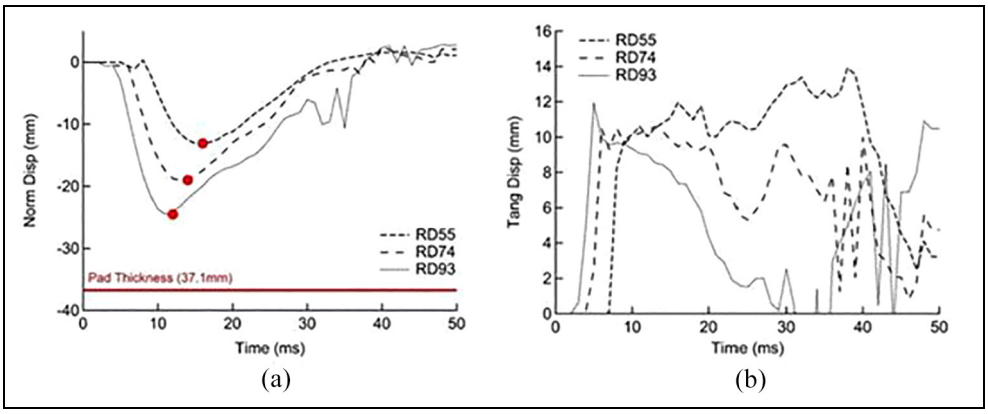

Figure 4. (a) Normal and (b) tangential displacement versus time data for Ridell rear (RD) impact at 5.5, 7.4, 9.3 m/s impact speed.

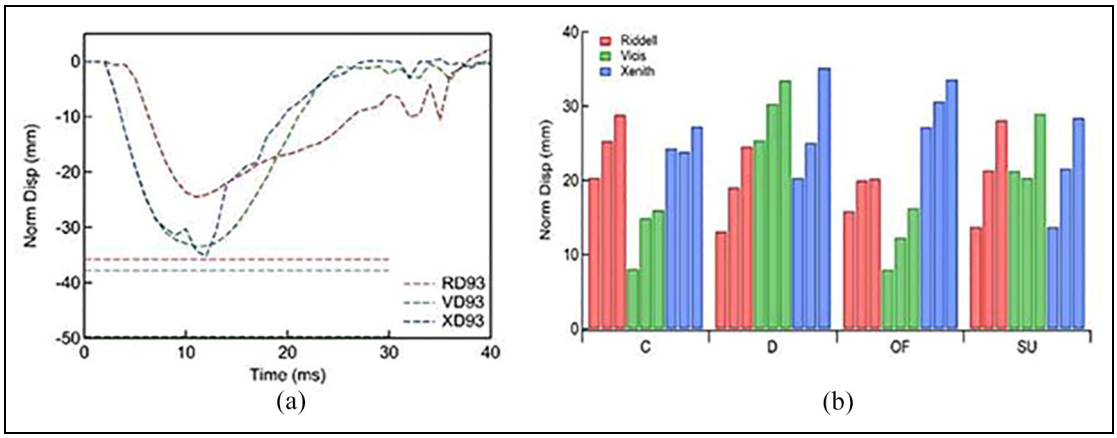

Figure 5. (a) Normal displacement data for all three helmets at fastest impact speed for the rear location and (b) peak normal displacement data for each helmet, speed, and location.

Results

Displacement time histories

Exemplary time history data for normal and tangential displacements from a Riddell rear impact are presented in Figure 3. Both curves showed a region of zero displacement from $t0$ to initial contact of the impactor with the helmet, followed by a rapid increase in displacement. The normal displacement versus time curve is roughly parabolic in shape, exhibiting more rapid loading to maximum displacement than unloading. A marker indicating the maximum pad thickness in the normal direction has been included in Figure 3(a) for reference. For this example, the pad compresses 37% of the total thickness upon impact. Tangential curves maintained a region of high displacement with a noisy response prior to the pads recovering.

Next, the influence of impact speed on displacement was investigated. When comparing time history curves for all three impact speeds for one helmet at one impact location, it was found that peak normal displacement increased as impact speed increased. Representative normal displacement time history data for the Riddell rear impact at all three speeds is presented in Figure 4(a). For these simulations, peak normal displacement was found to be approximately 12 mm at 15 ms for the slowest impact and 25 mm at 10 ms for the fastest impact, indicating a 108% increase in displacement. In general, these trends were observed for all impact speeds and locations except for the Vicis C and Xenith D locations.

Minimal change in tangential displacement was observed across impact speeds and clear trends were not established with regards to maximum displacement or rate. Representative tangential displacement time history data for the Riddell rear impact at all impact speeds is presented in Figure 4(b). In this chart, tangential displacements showed minimal change, ranging from 10 mm at the slowest impact to 12 mm at the fastest impact. There was no noticeable difference in the initial displacement rate.

The displacement time history profiles can be used to compare performance between helmets. A comparison of the three helmets impacted at the rear location at 9.3 m/s is presented in Figure 5(a). Vicis and Xenith helmets demonstrated nearly identical loading curves

Table 1. Correlation table for normal displacement versus helmet kinematics and impact severity metrics.

	HARM	HIC	DAMAGE	PRLA	PRAA	PRAV	Tang Disp
Norm Disp							
SU	0.79	0.80	0.48	0.79	0.40	0.61	0.24
C	0.23	0.19	0.29	0.20	0.51	0.41	0.06
OF	0.06	0.00	0.33	0.01	0.40	0.52	0.00
D	0.39	0.46	0.23	0.64	0.33	0.23	0.31
ALL	0.37	0.36	0.33	0.41	0.41	0.44	0.15

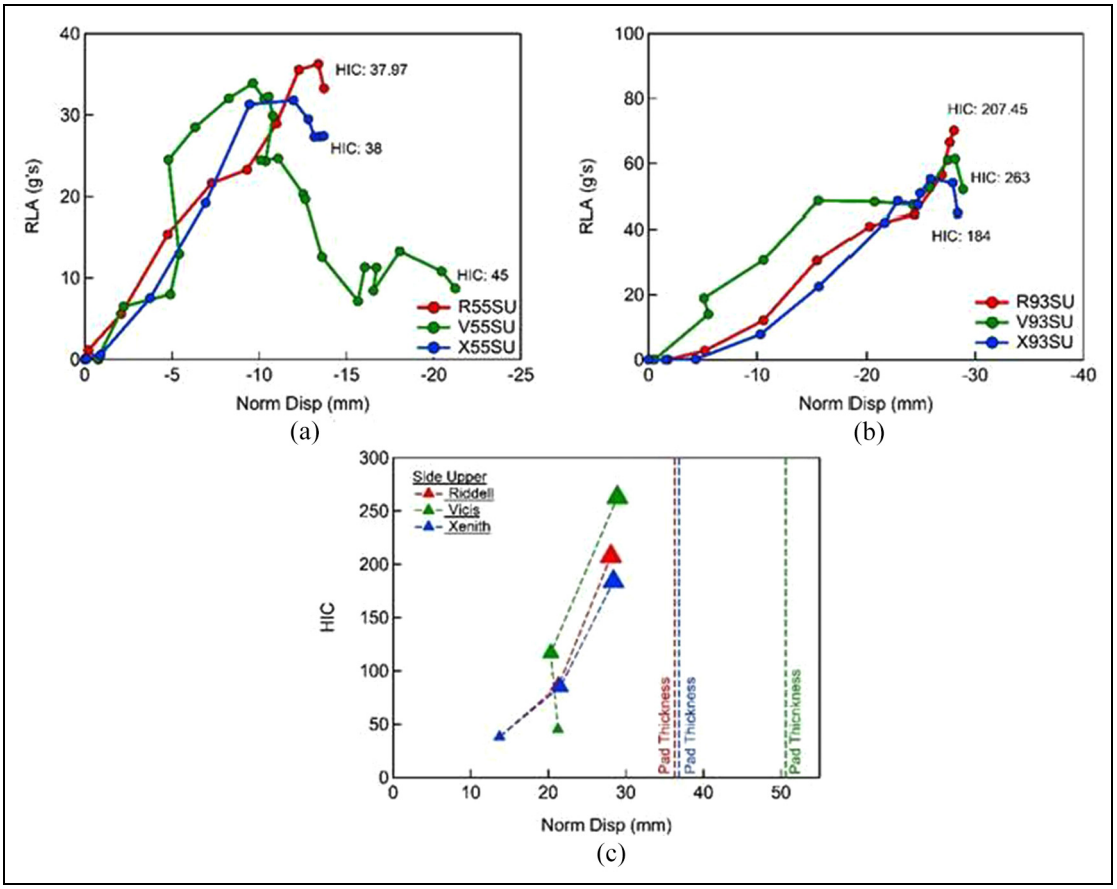

Figure 6. RLA versus normal displacement data for each helmet at the side upper (SU) location for: (a) 5.5 m/s, (b) 9.3 m/s, and (c) comparison of HIC versus maximum normal displacement for SU impacts for all impact speeds.

for normal displacement until reaching a maximum displacement of approximately 35 mm. The Riddell helmet underwent a noticeably slower displacement rate and lower displacement maximum. Once the maximum was achieved, the Riddell helmet demonstrated the slowest rate of recovery. A summary of peak displacement values for all impacts is presented in Figure 5(b). This chart can be used to compare peak deformations based on impact location, speed, and helmet model. For example, all three helmets show relatively similar values at the SU location, whereas the largest differences in performance occurred at the OF location. For example, the Xenith helmet experienced 12.82 and 17.37 mm more normal displacement than the Riddel and Vicis helmet at the OF location.

Correlations to normal displacement

Correlations between normal displacement and impact severity metrics and kinematic variables are presented in Table 1. Impact locations included in this analysis were C, D, OF, and SU. When comparing all impact locations collectively, normal displacement was not strongly correlated to any of the metrics ($R^2 = 0.33$–0.44) with the weakest correlation between normal and tangential displacement ($R^2 = 0.15$). Stronger correlations were observed when separating impacts by location. Most notably, the SU location demonstrated strong correlations between normal displacement and HARM, HIC, and PRLA ($R^2 = 0.79$–0.80). PRLA at the D location showed the next highest correlation of $R^2 = 0.64$. Virtually no correlation was

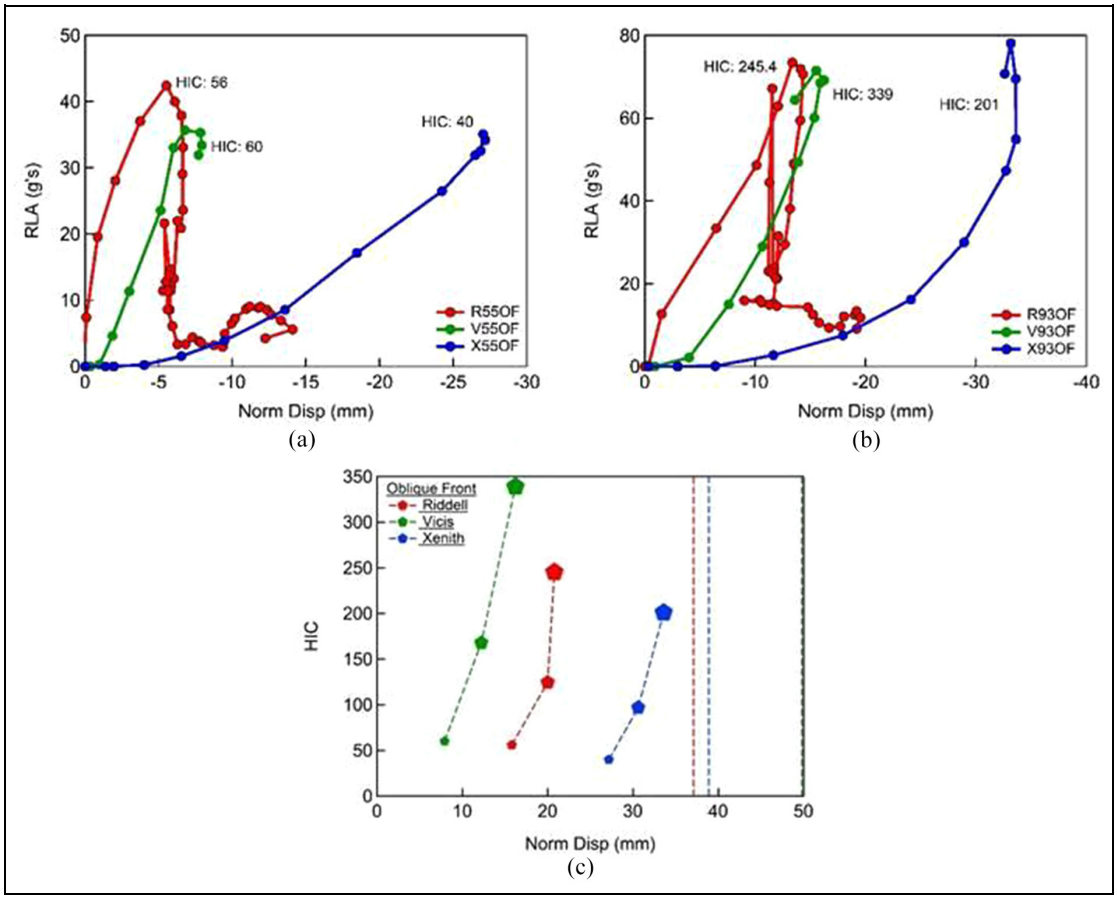

Figure 7. RLA versus normal displacement data for each helmet at oblique front location for: (a) 5.5 m/s, (b) 9.3 m/s, and (c) PRLA versus normal displacement for each helmet and speed at oblique front location.

found for the OF location when evaluating HARM, HIC, and PRLA.

For normal displacement versus PRLA, the strongest correlation was found at the SU ($R^2 = 0.79$) location. For the slowest impact speed, all three helmets experienced a maximum Resultant Linear Acceleration (RLA) between 32 and 37*g*, which occurred between 9 and 14 mm of normal displacement (Figure 6(a)). For the Riddell and Xenith helmets, the maximum normal displacement occurred shortly after PRLA. The Vicis helmet demonstrated continued displacement at much lower acceleration up to 22 mm. At the fastest impact speed tested, PRLA increased to 55.3–70.2*g*, and maximum displacements were between 27 and 29 mm (Figure 6(b)). The trend observed in all three speeds tested (7.4 m/s, not shown), was that RLA increased similarly with displacement for the Riddell and Xenith helmets. The Vicis simulations showed markedly higher RLA values leading up to PRLA. For example, at 10 mm of displacement at the 9.3 m/s impact, RLA was approximately three times higher for the Vicis helmet. When examining HIC scores for these impacts, the data is closely grouped per impact speed (Figure 6(c)). The one outlier is the Vicis impact at 5.5 m/s, which was shown to increase well past PRLA.

The lowest correlations between RLA and normal displacement were observed at the oblique front location. At 5.5 m/s impact speed, PRLA values ranged from 35 to 43*g* (Figure 7(a)). The Xenith helmet showed substantially more displacement than the other two helmets at this impact location. For example, PRLA was reached in the Riddell and Vicis helmets between 5 and 7 mm of displacement, while the Xenith helmet reached a PRLA at 27 mm. Similar trends were exhibited for all three helmets at the fastest impact speed (Figure 7(b)). The Riddell and Vicis helmets experienced similar peak accelerations (78 and 76*g*, respectively), however, the Ridell helmet showed a larger peak displacement (20 mm vs 13 mm), as the pad continued to deform well past PRLA. The Xenith helmet again showed a similar PRLA to the other two helmets but a peak displacement of 33 mm. For each impact speed, the Xenith helmet resulted in a markedly lower loading rate than did the other two helmets. This resulted in a lower HIC value for the Xenith helmet across impact speeds (Figure 7(c)).

Finally, the same analysis was employed for the rear location (Figure 8). Here, the Vicis and Xenith helmets exhibited a delayed loading response across all speeds. Both helmets deformed approximately 10 mm upon impact with no appreciable acceleration. After 10 mm,

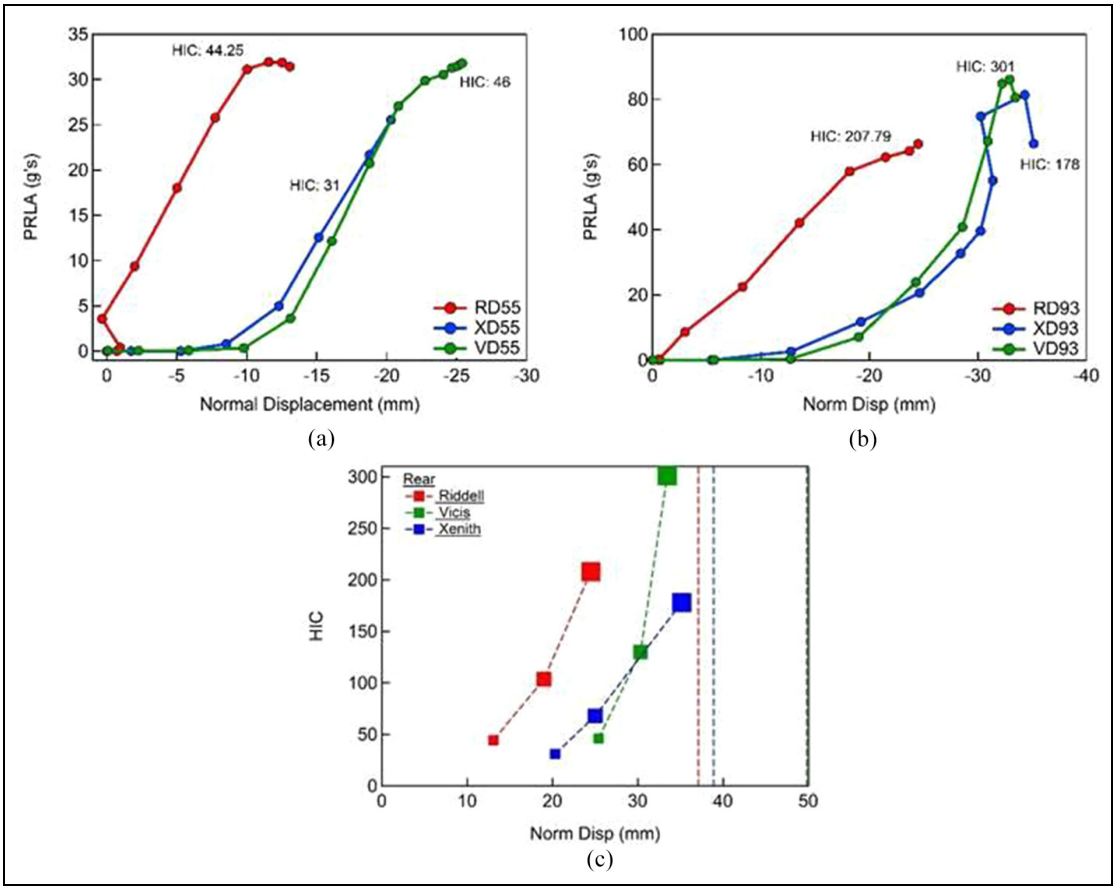

Figure 8. RLA versus normal displacement data for each helmet at rear location for: (a) 5.5 m/s, (b) 9.3 m/s, and (c) PRLA versus normal displacement for each helmet and speed at rear location.

Table 2. Correlation table for tangential displacement versus helmet kinematics and impact severity metrics.

	HARM	HIC	DAMAGE	PRLA	PRAA	PRAV	Norm Disp
Tang Disp							
SU	0.16	0.43	0.00	0.17	0.00	0.02	0.24
C	0.02	0.00	0.07	0.00	0.11	0.01	0.06
OF	0.00	0.00	0.01	0.00	0.10	0.00	0.00
D	0.27	0.06	0.41	0.23	0.65	0.32	0.31
ALL	0.11	0.12	0.12	0.10	0.22	0.09	0.15

both helmets experienced an increase in RLA proportional to displacement. Across impact speeds, the Riddell helmet showed the lowest peak displacement values, and the Vicis and Xenith helmets reached similar peak displacements. In addition, the Vicis helmet was found to have the highest HIC scores while the scores of Riddell and Xenith were within 17% difference (Figure 8(a) and (b)).

Tangential displacement time history evaluations

Tangential displacement was not found to correlate well with impact severity metrics or helmet kinematics. When including all locations, all R^2 values were determined to be < 0.2 except for tangential displacement versus Peak Resultant Angular Acceleration (PRAA; $R^2 = 0.22$). When separating by location, no correlations had an R^2 value > 0.45, except for tangential displacement versus PRAA at the D location ($R^2 = 0.65$). The worst correlated location was again the oblique front location ($R^2_{avg} = 0.02$; Table 2). The majority (61%) of correlations explored with respect to tangential displacement were < 0.1. Moreover, displacement was found to change by less than 20% across impact speeds except for Vicis C, SU, D, and Xenith D locations. Visual representation of peak displacement data as a function of helmet, location, and speed is presented in Figure 9.

Exemplary data for tangential displacement versus Resultant Angular Acceleration (RAA) is shown in

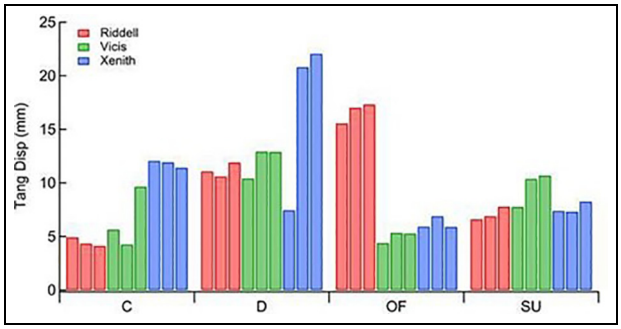

Figure 9. Peak tangential displacement data for all helmets, impact speeds, and locations used in this study.

Figure 10. For Riddell rear impacts, tangential displacement increased sharply to 10 mm at relatively low acceleration values. After 10 mm, the RAA increased dramatically with little change in displacement. This was common across the majority of impacts.

Peak tangential displacement evaluation

Evaluation of peak values for tangential displacement versus HIC indicated a relatively low change in displacement as impact speed increased. The only exception was observed in the Xenith helmet from 5.5 to 7.4 m/s impact speed at the rear location where tangential displacement was found to increase by approximately 14 mm and the PRAA increased by roughly 2000 rad/s^2. For the same comparison, the Riddell helmet resulted in a 2 mm change in displacement and a 1000 rad/s^2 increase in PRAA. The Vicis helmet experienced a change in displacement of roughly 3 mm and a similar increase in PRAA to the Riddell helmet (Figure 11(a)).

Assessment of tangential displacement versus Peak Resultant Angular Velocity (PRAV) at the side upper location resulted in trends similar to the previous data. Overall, the change in displacement as a function of impact speed was relatively low (> 3 mm). Each helmet experienced a roughly 10 rad/s increase in PRAV with increased impact speed. In general, impact speed was found to have a negligible effect on peak tangential displacement (Figure 11(b)).

Discussion

The purpose of this study was to investigate the relationship between components of helmet pad displacement and impact kinematics as well as the resulting impact severity metrics using computational modeling.

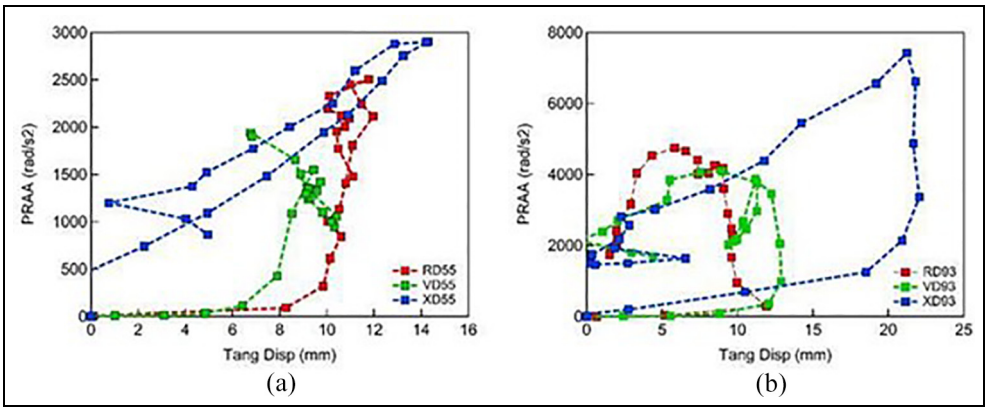

Figure 10. RAA versus tangential displacement data for each helmet at: (a) 5.5 m/s and (b) 9.3 m/s.

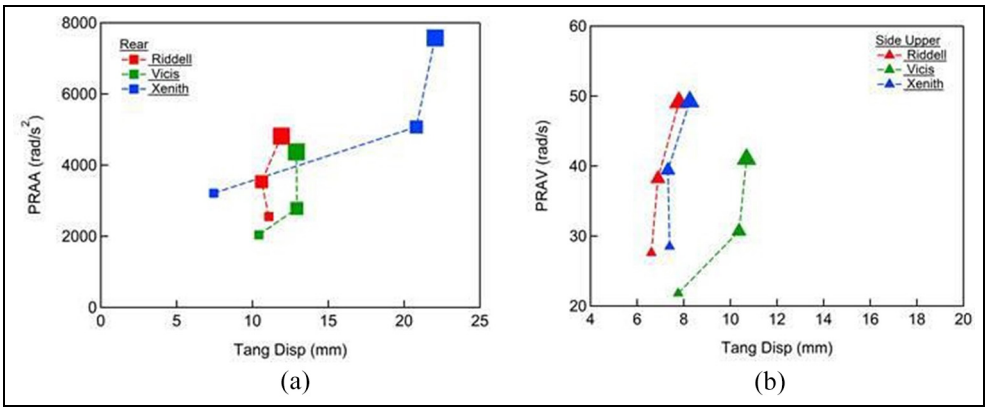

Figure 11. (a) Peak tangential displacement versus PRAA data for all helmets and speeds at the rear location and (b) peak tangential displacement versus PRAV data for all helmets and speeds at the side upper location.

A code was developed to separate the normal and tangential displacements of the helmet during simulated impact. We were able to successfully capture time history and peak values for both displacement values. To the best of our knowledge, no prior studies have evaluated the components of helmet pad deformation or related them to measures of impact severity.

We hypothesized that maximizing pad deformation would lower the corresponding impact severity metric. This notion comes from previous work by Camarillo et al.[21] who illustrated how optimal performance could be achieved if helmets maximize pad deformation without bottoming out. We looked for cases that would support this notion. For the SU location, it was shown that all three helmets had similar maximum normal displacements for each impact speed. For example, the maximum deformations increased by 32% for Riddell and Xenith helmets and 42% for the Vicis helmet from 7.4 to 9.3 m/s. As a result, a strong correlation was found between maximum normal displacement and HIC ($R^2 = 0.80$). Conversely, we examined the OF impact location where no correlation was present ($R^2 = 0.00$). Here, each helmet showed a dramatically different response. Vicis experienced the lowest displacements and highest HIC values, while the Xenith helmet showed the largest displacements and lowest HIC values. For all impact speeds, Xenith showed greater than two times the displacement compared to the Vicis helmet. As hypothesized, a clear trend could be seen that HIC decreased with increasing displacement for each impact speed. Our investigation would suggest that a strong correlation between displacement and impact metrics is an indicator that the helmets investigated are performing similarly for a given impact location. Conversely, the lack of any correlation is a result of dramatic differences in the deformation behavior of the helmets.

The OF location supported the notion that helmets that are best able to optimize their displacement across impact speeds will perform better under the impact conditions used. It is important to note here that a pad subjected to impact may become fully densified and will not continue to absorb energy; thus, simply maximizing displacement per impact may not always improve performance. For example, at this location and 9.3 m/s, the Xenith helmet compressed roughly 90% of the overall pad thickness. Increasing the impact speed likely would have resulted in the pad becoming fully densified, resulting in worse helmet performance.

The rear location presented some unusual behavior compared to other test locations. When comparing the Xenith to Riddell helmet, the results support the hypothesis that increased displacement for an impact will result in lower severity metrics. The Vicis helmet does not follow the same trend, experiencing the highest HIC value at all speeds despite having relatively high displacements. The first key here is that despite the high overall amounts of deformation, there is little increase in displacement with increased impact speed. For example, the Vicis helmet demonstrated only 3.15 mm of increased displacement from the 7.4 to 9.3 m/s speeds, whereas the Riddell and Xenith helmet demonstrated 5.5 and 10.13 mm, respectively. This may indicate the helmet liner is beginning to fully densify. Specifically, the Vicis helmet model used in this study incorporates many buckling columns between the headform and shell of the model. The gap between each of these columns is roughly equivalent to the diameter of the columns themselves. Thus, when buckled, the columns begin to collide and prevent each other from moving freely, essentially locking the helmet pad at that displacement. Therefore, while Figure 8(a) indicates a Vicis pad thickness of nearly 50 mm, the real usable space in which the helmet absorbs energy is likely closer to that of the other helmets. Because the space inside the helmet is smaller than the pad thickness, the pad densifies sooner than it appears, resulting in relatively high HIC values. The takeaway here is that the structure and underlying mechanisms of displacement are important factors in assessing impact severity.

The second thing to consider for the rear impact is that both Xenith and Vicis helmets demonstrate a large amount of displacement with very little initial acceleration (Figure 7(a) and (b)). Each helmet compresses approximately 10 mm or more before accelerations reach above 5*g*. This behavior was not seen in any of the other test locations. This may be a result of the headform shape in the simulation. The rear nape of the head is omitted in the model. As a result, the liner of a helmet may not perform as intended due to the missing surface area of the head. A previous study by Mills et al.[34] showed a high degree of non-uniform compression in the rear zone of the headform for simulated Riddell impacts. As a consequence, the displacement measurements at this location may not fully represent real-world conditions and alternative headform geometries may need to be considered.

In addition to the underlying structure and mechanics of an impact, it is imperative to understand how each severity metric is calculated. For example, HIC calculation involves maximizing the integral of acceleration over some specified time window for a given time history curve. Because of the caveat of maximizing this value, it is possible that impacts resulting in similar peak displacements and accelerations will have different HIC values; the shape of the acceleration time history curve greatly effects the resulting score. (An example of this can be seen in Figure 5(c), in which the Vicis helmet shows the highest HIC values with equivalent displacement values to other helmets.) A more detailed explanation of the influence of time history curve shape and impact severity scores has been given previously by Chatham et al.[35]

Little variation in tangential displacement was seen in each helmet with increasing impact speed (Figures 9 and 11). As a result, tangential correlations between displacement and severity metrics were weak (Table 2). However, R^2 value alone fails to fully capture the

mechanics of the impact. Figure 10 shows that much of the rotational displacement was experienced without any significant angular acceleration. In Figure 11(a), we can see that both the Riddell and Vicis helmets had minimal change in displacement as a function of speed while PRAA increased. On the other hand, the Xenith helmet shows a large jump in displacement from 5.5 to 7.4 m/s with minimal change from 7.4 to 9.3 m/s.

Here, the pad under impact is only partially densified at 5.5 m/s. At the next speed, the impact has enough energy to displace the pad near the point of full densification. It follows then that the helmet should see a much steeper slope from 7.4 to 9.3 m/s, as it was nearly densified at the middle impact speed (Figure 11). In short, regression analysis of these data sets fails to accurately describe the mechanics of impact, and the means by which the pad displaces has a significant role in the resulting helmet kinematics. It is imperative to understand how the helmet is moving to interpret the influence of displacement on impact severity metrics.

There are a few limitations identified within this experiment which may help guide future work toward a similar goal. First, the data is limited by the number of impacts specified in the testing protocol. For any given helmet and impact location, only three peak data points occur which is too small a data set to accurately assess correlations. Future work should seek to expand the data set acquired here by testing helmets at a greater range of speeds. The next limitation is that we did not explore the upper and lower bounds of performance for these helmets. By testing at a range of speeds which approach 0 m/s as well as 20 + m/s researchers will gain insight into the limits of individual helmets and better be able to correlate displacement to impact metrics.

The next limitation of this study is that these helmet models have now aged several years and have mostly been replaced by newer versions across most levels of play. Moreover, there are certain instances in which the models could be further refined to prevent excessive element distortion or solid object penetration, which may influence simulation results. Overall, the validated models used in this study are a reasonable approximation of real-world impacts, and they provide an opportunity to investigate the effects of individual helmet components, including energy-dissipating pads, on final values of impact severity metrics.

The final limitation of this study relates to the code used in the calculation of tangential displacement. Tangential displacements are all calculated as resultant, and therefore have no indication of direction. However, previous studies have indicated that direction and axis of rotation play a significant role in the calculation of impact severity metrics.[32] Future work should seek to determine the specific direction of displacement and evaluate correlations to the associated impact severity metrics and helmet kinematics.

In short, while helmet impacts in some locations showed similar performance, other locations exhibited variance in deformation mechanics and impact severity scores. This research has provided a means of describing the relationship between pad displacement and helmet impact severity metrics in certain impact locations. For example, strong correlations were found between normal displacement and impact severity metrics at the SU location, the highest weighted impact location in calculating HPS scores. In addition, this research has uncovered some limitations in FE analysis of football helmet impacts. This includes FE model definitions which could benefit from improved refinement and a limited number of data points specified in the testing protocol. Additional limitations include metric calculations which may not utilize the full-time history of an impact or may be unitless and not describe impact mechanics.

Moreover, the relationship between impact speed, pad deformation, and impact severity metrics is generally well understood when considering simpler impact scenarios such as a drop test. However, to the best of our knowledge there has yet to be a study which has investigated these relationships in a helmet system where the effects of multiple components and non-uniform geometries contribute to the overall response. Finally, the researchers realize that certain helmet models (e.g. Vicis) may not adhere to a more standard compressive response of the pads. While this particular helmet model uses buckling columns in its design, these columns possess a bending stiffness and may buckle onto themselves preventing further displacement. Thus, the researchers have chosen to keep each of the models studied in this analysis as understanding of impact mechanics and pad deformation will be useful information to future engineers or product developers seeking to iterate through new helmet pad designs.

Conclusion

In this study we developed a method that is able to explicate the components of displacement from an impact and allows a new means of assessing the performance of football helmets. When evaluated by location, some relatively strong correlations arise with respect to normal displacement; however, this does not occur with tangential displacement. When comparing equivalent impact speeds, an increase in normal displacement typically leads to a lower HIC value. This indicates that helmets which are most effectively able to utilize the thickness of the pad under impact will perform better and experience lower severity metrics. However, this relationship was location dependent, with a hit from a side to the upper portion of the helmet demonstrating a strong correlation, and an oblique hit to the front of the helmet demonstrating a weak correlation. In addition, tangential displacement was found to be relatively unaffected by impact speed, producing similar displacement values at all impact speeds. All impacts simulated in this study orient the impactor roughly normal to the

shell and internal pads; thus, it follows that normal displacements are expected to be more heavily influenced in these types of impacts. Overall, our results suggest that the relationship between pad displacement and head impact severity metrics is highly dependent on impact location.

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Supplemental material

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